OSPF Network Configuration Lab Report

1. Objective

To implement OSPF routing in a multi-router topology, verify neighbor adjacencies, and troubleshoot connectivity issues to ensure full routing table convergence and end-to-end connectivity.

2. Topology Overview

The network consists of Area 0 with five routers (R1, R2, R3, R4, R5) and an ISP connection at R5. Multiple interface types are utilized, including Serial (Point-to-Point) and GigabitEthernet (Broadcast segments).

3. Configuration Summary

OSPF was configured using the `router ospf 1` process. Key configurations included:

- Advertising networks using wildcard masks in Area 0.
- Defining default route propagation on the edge router (R5) using `default-information originate`.
- Adjusting interface OSPF behaviors (e.g., ensuring correct Network Types).

4. Troubleshooting Process

Issue 1: OSPF Network Type Mismatch

On R3, the GigabitEthernet interface initially defaulted to Broadcast, leading to unnecessary DR/BDR election. When modified to Point-to-Point, connectivity was restored.

Issue 2: Connectivity to Internet (8.8.8.8)

PC1 could not reach the internet. Troubleshooting revealed that while R5 had a default route, it was not being propagated into the OSPF domain. The `default-information originate` command was applied to R5, allowing other routers to learn the path to the ISP.

5. Key Commands

```
! To enable OSPF
router ospf 1
network [network_id] [wildcard_mask] area 0

! To originate default route (on Edge Router)
default-information originate

! Verification commands
show ip ospf neighbor
show ip route
show ip ospf database
```

6. Final Results

The routing tables are converged, with OSPF routes marked with 'O' and default routes marked as 'O*E2'. Ping tests confirm end-to-end connectivity between subnets and the internet.